

Yaohui Shu

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Ph.D. candidate in Applied Mathematics and Statistics working across computational applied mathematics, stochastic modeling, information-theoretic causal inference and turbulent processes in complex systems. The dissertation develops entropy-based methods that identify cause and effect, verified on physical models of turbulence with known causal structure.

SKILLS

Causal inference: causal graph learning (PC, FCI, PCMCI), transfer entropy, information flow, conditional mutual information

Modeling and simulation: stochastic differential equations (Langevin, Ornstein-Uhlenbeck), Fokker-Planck equations, Itô calculus, Girsanov's theorem, maximum entropy, Monte Carlo with convergence testing, finite-volume PDE solvers

Statistics and ML: time-series analysis, change-point detection, latent-factor models (PCA, ICA), hypothesis testing with surrogate and permutation nulls, bootstrap, false-discovery control, gradient boosting, LSTM networks

Programming: Python (NumPy, SciPy, pandas, scikit-learn), C++, MATLAB, R; Git, Linux, SLURM/HPC

EDUCATION

Stony Brook University, Ph.D., Applied Mathematics and Statistics, GPA 3.78/4.0 Expected May 2027

Dissertation: *Uncovering Causal Mechanisms of Aerosol-Cloud-Turbulence Interactions Using Physics- and Data-Driven Methods*

University of Rochester, M.S., Finance, GPA 3.81/4.0; Certificate in Quantitative Finance Aug 2021 – Aug 2022

B.A., Finance, Zhongnan University of Economics and Law (2021); B.S.B.A., University of Rhode Island (2020, GPA 3.98)

RESEARCH EXPERIENCE

Stony Brook University Stony Brook, NY

Doctoral Researcher, Causal Inference for Complex Systems

Aug 2023 – Present

- Developing a causal discovery method (partial ancestral graphs) that handles feedback loops and unmeasured drivers; standard methods assume neither exists, so they can report false cause-and-effect links.
- Proposed a measure of causal strength that stays valid under feedback, derived in closed form with Girsanov's theorem.
- Built a Python package that brings entropy-based measures into causal graph learning, with nearest-neighbor independence tests, surrogate-data calibration and false-discovery control, checked against closed-form results.
- Early-stage application to financial time series: hidden-driver graph search, regime detection, event studies.

Brookhaven National Laboratory (U.S. Department of Energy)

Upton, NY

Graduate Research Assistant, Stochastic Modeling and Information Flow

Jan 2024 – Jan 2026

- Primary developer of a DOE-funded cloud turbulence project: built its theory, simulations and code.
- Extended a Brownian-motion droplet-growth model, a stochastic dynamical system, to fluctuations with memory and quantified its causal strength with information flow from information theory (first author, *Geophysical Research Letters*).
- Reconciled two competing theories of droplet size distributions, maximum entropy and Langevin dynamics, and analyzed the stochastic system under different physical constraints.
- Derived exact size distributions for a Fokker-Planck model, solved the 2D case with a finite-volume solver, and showed through convergence studies that the boundary condition sets the spread more than any physical parameter.

Graduate Teaching Assistant, AMS 595 Fundamentals of Computing (Python, C++, MATLAB)

Aug 2026 – Present

PROJECTS

Entropy-Integrated Causal Graph Learning (GitHub)

Working draft

- Replaces correlation and Gaussian assumptions in causal graph learning with entropy-based measures: transfer entropy to initialize the graph, conditional mutual information tests to prune it, and entropy-regularized NOTEARS.
- Estimates the strength of each recovered link with information flow where the system is close to linear and with doubly robust machine learning (DML) elsewhere, to be validated on stochastic systems with known causal links.

Pretrained Causal Discovery for Time Series

Working draft

- A transformer pretrained on simulated systems with known causal graphs that outputs a causal graph with a confidence score on every link, aimed at data with hidden drivers, feedback and regime changes.
- Built so far: simulators with known ground truth, scoring metrics and classical baselines (LPCMCI, PCMCI+).

PUBLICATIONS

- **Shu, Y.**, Liu, Y., Zhang, T., McGraw, R., Li, X. (2026). Stochastic condensation with information flow for causality analysis. *Geophysical Research Letters*, 53(17), e2026GL123056. Code: [GitHub](#)
- In preparation: *Physical Review E; Journal of the Atmospheric Sciences*; causal discovery methodology paper.
- Presentations: AGU Fall Meeting 2025 (first-author poster); 106th AMS Annual Meeting 2026 (co-author).